

```

1  import numpy as np
2
3  a, b, n = 1, 3, 10
4  tau = (a + b) / 2 # true mean = 2
5  np.random.seed(0)
6  # Number of simulations (increase for more accuracy)
7  N = 200000
8  squared_errors = []
9
10 for _ in range(N):
11     # 1) simulate a sample of size n from Uniform(a,b)
12     x = np.random.uniform(a, b, size=n)
13     # 2) compute MLEs of a and b: min and max
14     a_hat = np.min(x)
15     b_hat = np.max(x)
16     # 3) MLE for tau = (a+b)/2 is (a_hat+b_hat)/2
17     tau_hat = (a_hat + b_hat) / 2
18     # 4) squared error
19     squared_errors.append((tau_hat - tau)**2)
20
21 # Monte Carlo estimate of MSE
22 mse_hat = sum(squared_errors) / N
23 print("Estimated MSE of MLE tau_hat:", mse_hat)
24
25 # (Optional) plug-in estimator is sample mean; its MSE here equals Var(X)/n = ((b-a)^2/12)/n
26 mse_bar = ((b - a)**2 / 12) / n
27 print("MSE of sample mean (analytic):", mse_bar)

```

Estimated MSE of MLE tau_hat: 0.015123777280892545
 MSE of sample mean (analytic): 0.03333333333333333